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JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE NAME

CLASS

BIOLOGY **8875/01**

Paper 1 Multiple Choice **20 September 2013**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and class on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

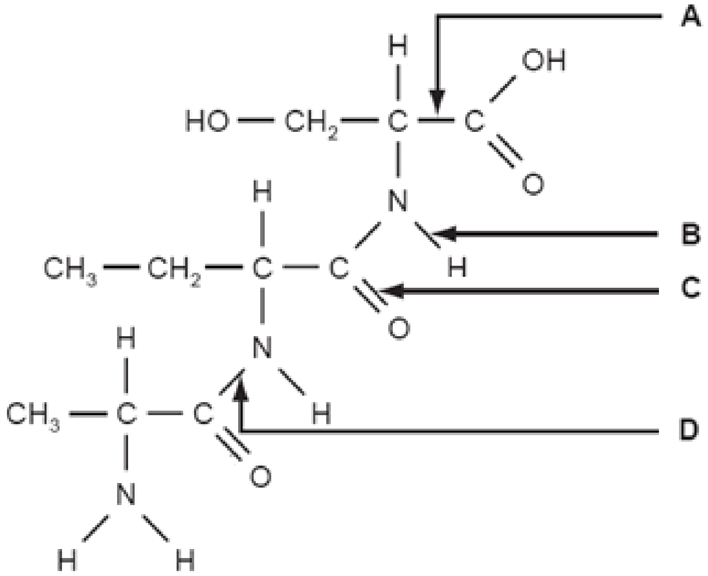
This document consists of **13** printed pages.

[Turn over

1 Which row correctly identifies all the locations of ribosomes in a eukaryotic cell?

	free in cytoplasm	in mitochondria	attached to endoplasmic reticulum	attached to nuclear envelope	in nucleus
A	✓	✗	✓	✗	✗
B	✓	✓	✓	✗	✗
C	✗	✓	✓	✗	✗
D	✗	✗	✓	✓	✓

2 The diagram shows a molecule. Which arrow labels a peptide bond?



3 What are the features of triglycerides?

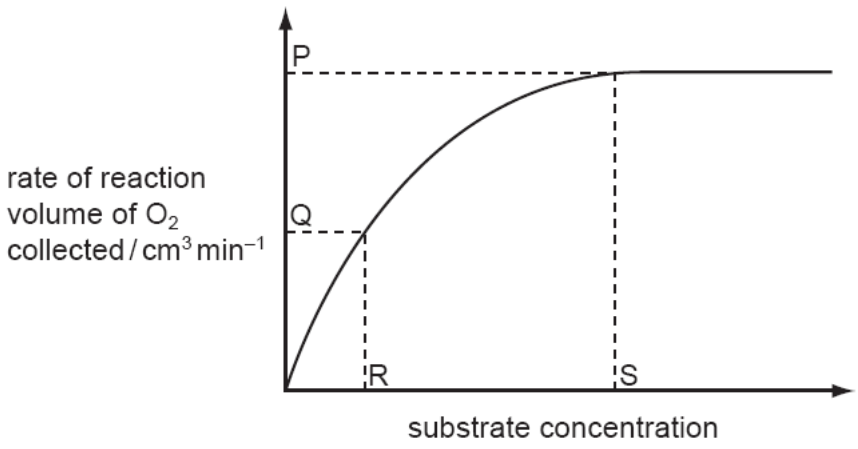
	polar	less dense than water	higher energy value than carbohydrates	lower proportion of hydrogen than in carbohydrates
A	✓	✓	✗	✗
B	✓	✗	✓	✓
C	✗	✓	✓	✗
D	✗	✗	✗	✓

4 Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time.

Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are correct?

1 At P, the rate of reaction is limited by the concentration of enzyme.
2 At Q, all of the enzyme active sites are occupied by substrate molecules.
3 At Q, the rate of reaction is limited by the concentration of the substrate.
4 R represents K_m where the reaction rate = V_{max} / 2.
5 At S, all of the enzyme active sites are occupied by substrate molecules.

A 2 and 3 only **B** 1, 2, 3 and 4 only **C** 1, 2, 4 and 5 only **D** 1, 3, 4 and 5 only

5 Which statement is incorrect for mitotic cell division?

A DNA is replicated semi-conservatively during mitosis.
B DNA is normally unchanged from one generation of cells to the next.
C The daughter cells have the potential to produce the same enzymes as the parent cell.
D The same quantity of DNA is distributed to the nuclei of two new cells.

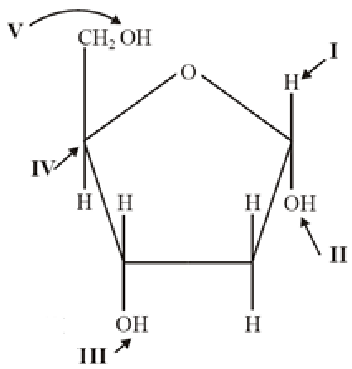
6 For organisms undergoing sexual reproduction, a reduction division occurs before fertilisation.

Which reason(s) explain why this is necessary?

1 increase genetic variation
2 prevent doubling of the chromosome number
3 reduce the chances of mutation

A 1 only **B** 2 only **C** 1 and 2 only **D** 1, 2 and 3

7 To which parts of the deoxyribose molecule do phosphates bind in DNA?

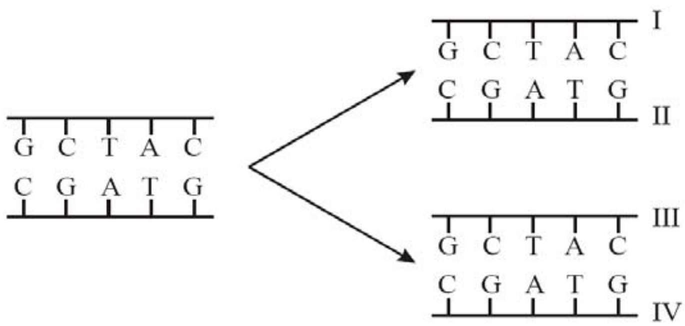


A I and V **B** III and IV **C** II and III **D** III and V

8 Which features of DNA enable it to meet these requirements as a molecule of inheritance?

	requirement of DNA molecule			
	ability to remain stable	ability to contain information	ability to transfer information	ability to replicate
A	complementary base pairing	formation of mRNA for translation	sequences of nucleotides	sugar-phosphate backbone
B	formation of mRNA for translation	complementary base pairing	sugar-phosphate backbone	sequences of nucleotides
C	sequences of nucleotides	sugar-phosphate backbone	complementary base pairing	formation of mRNA for translation
D	sugar-phosphate backbone	sequences of nucleotides	formation of mRNA for translation	complementary base pairing

9 The diagram shows a short section of DNA molecule before and after replication. If the nucleotides used to replicate the DNA were radioactive, which strands in the replicated molecules would be radioactive?



A I and II only **B** I and III only **C** II and III only **D** I, II, III and IV

10 Which statements about the genetic code are correct?

1 The genetic code has redundancy and is degenerate.
2 There is only one codon for the amino acid methionine.
3 Codons act as 'stop' and 'start' signals during transcription and translation.
4 Prokaryotes generally use the same genetic code as eukaryotes.

A 1 and 2 only **B** 1, 2 and 3 only **C** 1, 2 and 4 only **D** 1, 2, 3 and 4

11 5' – CAU – 3' is a codon in mRNA that specifies the amino acid histidine (His) in position 58 in the α chain of haemoglobin. What is the corresponding anti-codon in tRNA?

A ATG
B AUG
C GUA
D GTA

12 Base substitution mutations can have the following molecular consequence except

A changes one codon for an amino acid into another codon for that same amino acid.
B codon for one amino acid is changed into a codon of another amino acid.
C reading frame changes downstream of the mutant site.
D codon for one amino acid is changed into a translation termination codon.

13 A parent organism of unknown genotype is mated in a test cross. Half of the offspring have the same phenotype as the parent.

What can be concluded from this result?

A The parent is heterozygous for the trait.
B The parent is homozygous dominant for the trait.
C The parent is homozygous recessive for the trait.
D The trait being inherited is polygenic.

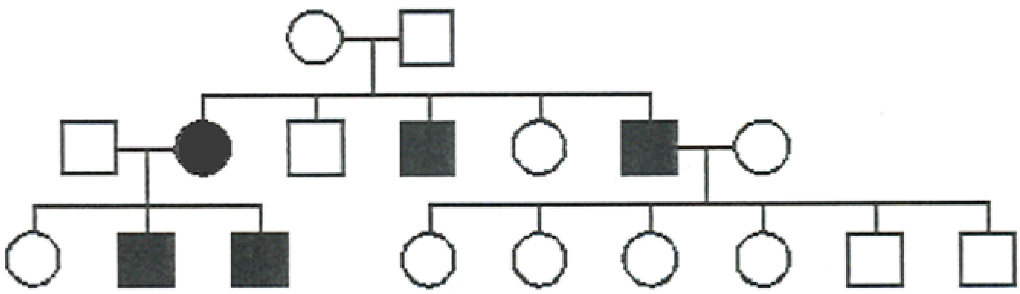
14 The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

A one gene with a pair of co-dominant alleles
B one gene with multiple alleles
C sex linkage of the allele for grey coat colour
D two genes, each with a dominant and recessive allele

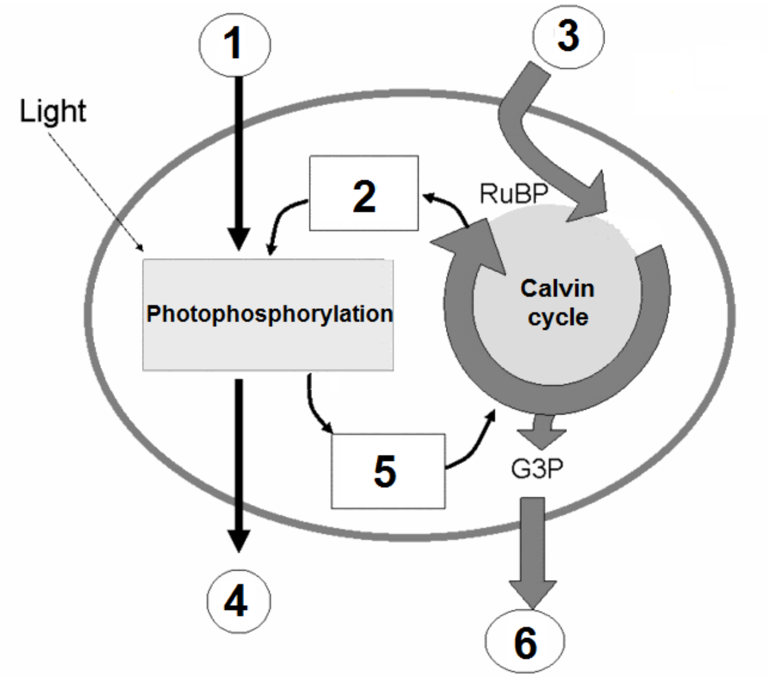
15 The following pedigree depicts the inheritance of a rare hereditary disease affecting muscles.



What is the mode of inheritance of this disease?

A autosomal dominant
B autosomal recessive
C X-linked dominant
D X-linked recessive

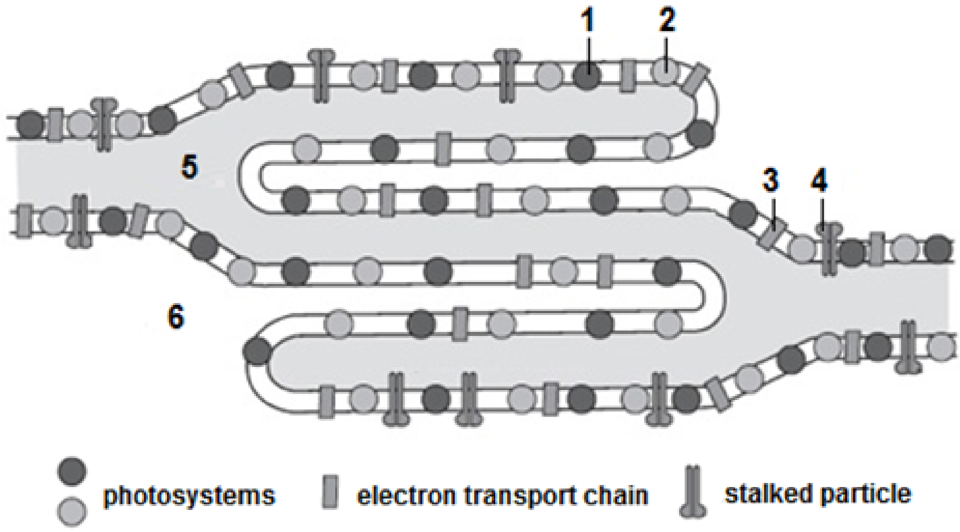
16 The diagram summarises the reactions during photosynthesis.



Which of the following correctly identifies the substances involved?

	1	2	3	4	5	6
A	H_2O	ADP, NAD	CO_2	O_2	ATP, NADH	glucose
B	O_2	ADP, NADP	glucose	H_2O	ATP, NADPH	CO_2
C	H_2O	ATP, NADPH	CO_2	O_2	ADP, NADP	glucose
D	H_2O	ADP, NADP	CO_2	O_2	ATP, NADPH	glucose

17 The figure shows the arrangement of photosystems, protein complexes containing chlorophyll molecules, on the thylakoid membrane of a plant chloroplast.



Which row correctly identifies the structure or function of the labelled regions?

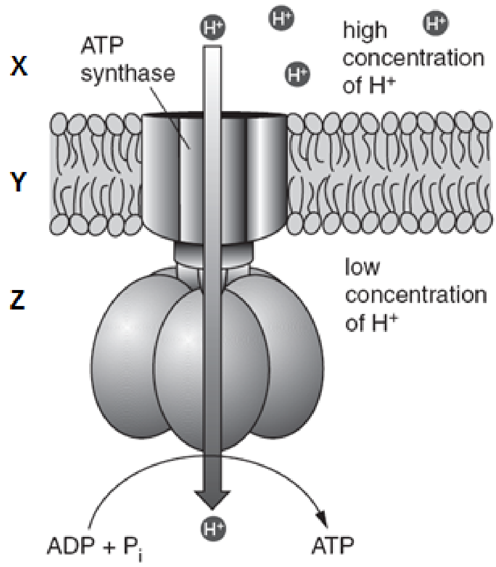
	site of photoactivation of chlorophyll	site of photolysis of water	site of ATP synthesis	stroma	thylakoid lumen
A	1 and 2	1 and 2	4	6	5
B	1 and 2	1	4	6	5
C	1, 2 and 3	1	3 and 4	5	6
D	1, 2 and 3	1 and 2	3 and 4	6	5

18 During substrate-level phosphorylation, ATP is synthesised from ADP and inorganic phosphate.

What is the immediate source of energy for this reaction?

A chemical bond energy released during the light independent stage of photosynthesis
B chemical bond energy released during glycolysis and the Krebs cycle
C potential energy of protons diffusing through the inner mitochondrial membranes in mitochondria
D potential energy of protons diffusing through thylakoid membranes in chloroplasts

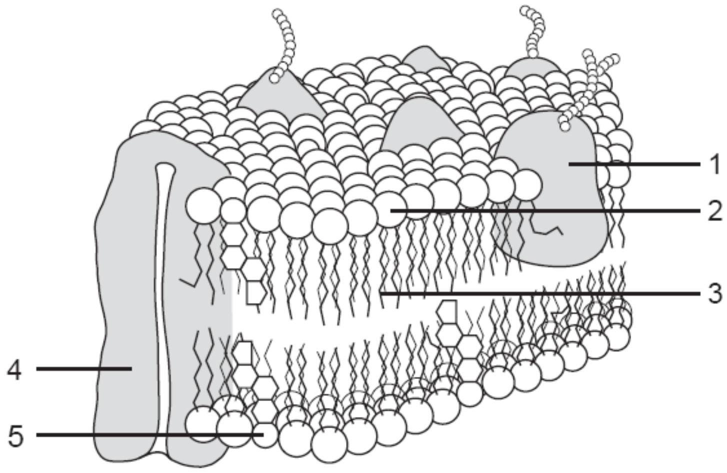
19 The diagram shows part of a membrane in a cell.



Which of these descriptions is true?

A X is the intermembrane space, Y is the inner mitochondrial membrane and the diagram shows ATP synthesis in a chloroplast.
B Y is the inner mitochondrial membrane, Z is the cytosol and the diagram shows ATP synthesis in a mitochondrion.
C X is the intermembrane space, Y is the inner mitochondrial membrane and the diagram shows ATP synthesis in a mitochondrion.
D Z is the intermembrane space, X is the matrix and the diagram shows ATP synthesis in a mitochondrion.

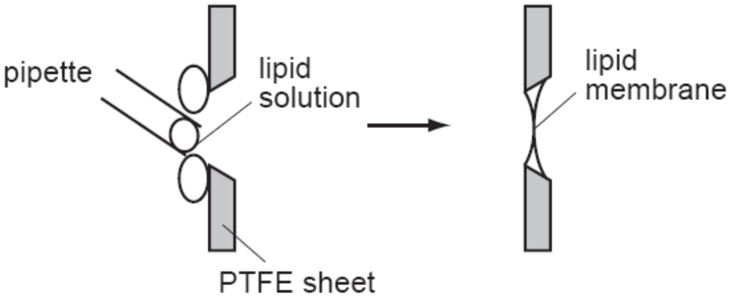
20 The diagram shows part of the cell surface membrane.



Which components help to maintain the fluidity of the membrane?

A 1 and 3 only **B** 1 and 4 only **C** 2 and 5 only **D** 3 and 5 only

21 Lipid membranes can be formed in the laboratory by painting phospholipids over a PTFE sheet with a hole in it.



Such a lipid membrane is impermeable to water-soluble materials including charged ions such as Na^+ or K^+.

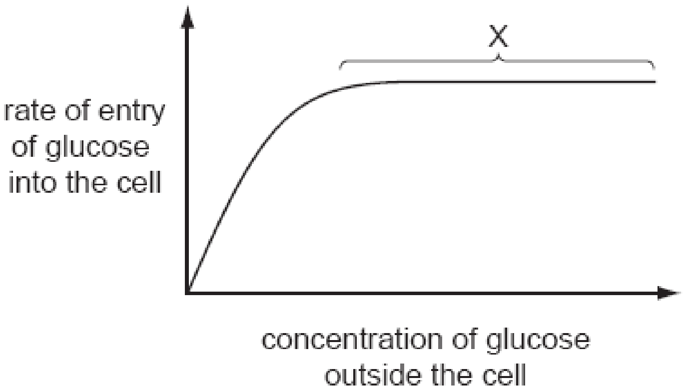
In one experiment with Na^+ ions, current flow (flow of charged ions) was investigated. No current flowed across the membrane until a substance called gramicidin was added, at which time current flowed.

Which statement is consistent with this information and your knowledge of membrane structure?

Gramicidin becomes incorporated into the membrane and is

A a carbohydrate molecule found only on the outside of the membrane.
B a non-polar lipid which passes all the way through the membrane.
C a protein molecule with both hydrophilic and hydrophobic regions.
D a protein molecule which has only hydrophobic regions.

22 The graph shows how the rate of entry of glucose into a cell changes as the concentration of glucose outside the cell changes.



What is the cause of the plateau at X?

A All the carrier proteins are saturated with glucose.
B The carrier proteins are denatured and no longer able to function.
C The cell has used up its supply of ATP.
D The concentrations of glucose inside and outside the cell are equal.

23 Which two features contribute to the great tensile strength of cellulose?

1 glycosidic bonds linking the long chains of 1,4 α-glucose molecules
2 the -OH groups of the glucose molecules project outwards and form H bonds with neighbouring chains
3 the strength of the glycosidic bonds between the neighbouring chains of molecules
4 the successive glucose molecules are orientated at 180° to each other

A 1 and 3 only **B** 1 and 4 only **C** 2 and 3 only **D** 2 and 4 only

24 Which statement(s) is/are acceptable parts of Darwinian evolutionary theory?

1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
2 In competition for survival, the more aggressive animals are more likely to survive.
3 Species perfectly adapted to a stable environment will continue to evolve.
4 Variation between individuals of a species is essential for evolutionary change.

A 4 only
B 2 and 3 only
C 3 and 4 only
D 1, 2 and 4 only

25 Before the settlement of California in the 1800s, the elk population was very large. By about 1900 there were only a few dozen elk left.

Owing to protection, there are now about 3000 elk living in a small number of isolated herds.

Unfortunately, some of the elk in all the herds have difficulty grazing due to a shortened lower jaw.

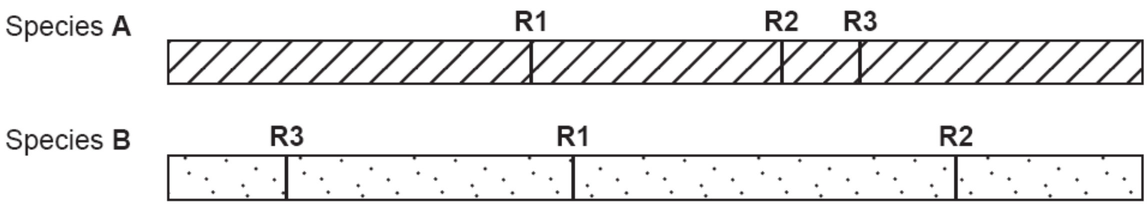
Which statement(s) is/are correct?

1 The early settlers only hunted elk that could graze.
2 There was a mutation affecting jaw size in one of the herds.
3 There is random mating within each herd.
4 There was directional selection favouring short jaws.

A 3 and only
B 1 and 2 only
C 2 and 4 only
D 2, 3 and 4 only

26 Restriction enzymes can be used to cut DNA at specific sites. Genes such as the gene for insulin can be cut from the chromosome of one species and as a result of ligation joined to the chromosome of another species forming recombinant (hybrid) DNA.

The figure shows two chromosomes from different species. The specific restriction enzyme sites (**R1**, **R2**, and **R3**) are shown.



Both chromosomes were cut at restriction site **R1** with *Eco*RI which recognises the sequence GAATTC and produces restriction fragments with sticky ends. The fragments were mixed and allowed to join to form recombinant (hybrid) DNA.

What is the maximum number of ways in which the fragments can join?

A 2 **B** 4 **C** 6 **D** 8

27 In which way(s) is/are the polymerase chain reaction (PCR) similar to the replication of DNA?

1 DNA is heated to break hydrogen bonds
2 DNA unwinds
3 RNA primers are used
4 DNA polymerase are required

A 2 only **B** 2 and 4 only **C** 1 and 3 only **D** 2, 3 and 4

28 Some of the steps involved in DNA analysis are listed below:

1 transfer segments of DNA to nitrocellulose membrane
2 extraction of DNA
3 gel electrophoresis
4 treating DNA with restriction enzymes
5 autoradiography
6 hybridise with probe

The correct sequence is

A 2 → 4 → 3 → 1 → 6 → 5
B 4 → 2 → 3 → 1 → 6 → 5
C 4 → 2 → 1 → 3 → 6 → 5
D 2 → 4 → 3 → 1 → 5 → 6

29 A key feature of most multicellular organisms is the ability to differentiate and produce specialised cells.

Which row best describes the ability of zygotic cells to differentiate?

	totipotent	pluripotent	multipotent
A	✓	✓	✓
B	✓	✗	✓
C	✓	✗	✗
D	✗	✓	✓

Key: ✓ = ability, ✗ = no ability

30 Which row correctly states the effects and possible consequences of producing the various genetically-modified organisms?

	Bt corn		GM salmon		Golden rice	
	effect	possible consequence	effect	possible consequence	effect	possible consequence
A	improve quality	might become weeds	improve yield	displace native species	improve yield	might become weeds
B	improve yield	toxic to closely related species	improve yield	displace native species	improve quality	cause allergies
C	improve yield	might become weeds	improve quality	cause allergies	improve yield	cause allergies
D	improve yield	toxic to closely related species	improve yield	cause allergies	improve quality	might become weeds